Blockchain & DeFi Datas

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Plan

- Blockchain data: volumes, types, economic value
- Levels of data aggregation: from raw to AI
- Raw data and RPC tools
- Indexers: The Graph, Goldsky, Bitquery, Space and Time
- Aggregators: Etherscan, Dune, Defillama, Dexscreener, GeckoTerminal
- Market data and Oracles
- AI $\times$ Blockchain Data: use cases

Why Blockchain Data is Hard

Blockchain is a transparent and decentralized database by design — every block, every transaction, every event is public. But answering even simple practical questions is surprisingly difficult:

- How many actual users does a DeFi protocol have?
- What is the true TVL of a multichain protocol without double-counting?
- How do we measure the real trading volume of a DEX pair?
- Is this new token a scam, or does it have legitimate holders?

Blockchain Sizes

Approximate full archive sizes:

- Bitcoin: $\sim$0.6 TB
- Ethereum: $\sim$20 TB (archive)
- Solana: $\sim$400 TB
- BSC: $\sim$2.5 TB
- Tron: $\sim$0.4 TB
- Other chains: $\sim$40 TB
- **Total: $\sim$450 TB and growing**

Node types:

- Light – headers only
- Full – recent state + all blocks
- Archive – full history of all states

Data Types and Who Pays for Them

Main data categories:

- **Transactions** – transfers, calls
- **Events** – emitted by contracts
- **State** – balances, storage
- **Market data** – prices, volumes, order books
- **DeFi / Tokens** – TVL, APY, liquidity
- **Wallets** – flows, labels, clusters
- **Network data** – mempool, validators, gas

Economic value – who pays:

- Traders → market data, low latency
- Protocols → events, on-chain analytics
- Compliance / forensics → wallet flows (Chainalysis, TRM)
- Researchers → everything, cheap
- Agents / bots → real-time streams

Levels of Aggregation

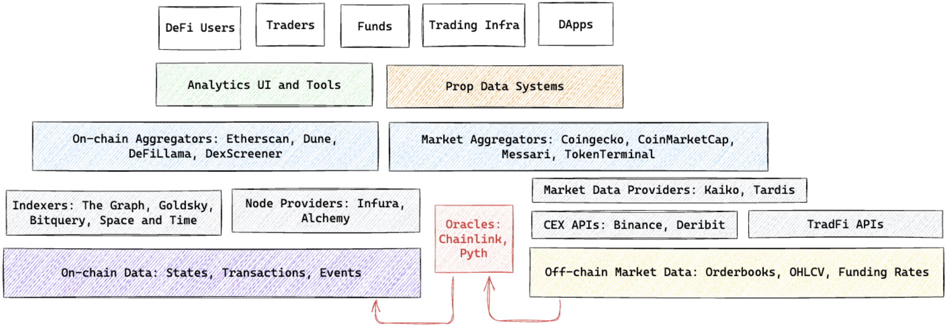


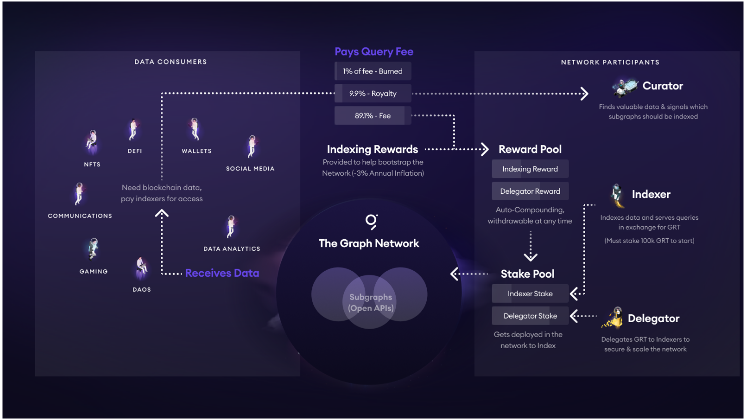
Figure 1: Data Levels

Raw Data

- Run your own node: Geth, Erigon, Reth (Ethereum); Solana validator
- RPC / Node providers: Alchemy, Infura, QuickNode, Ankr, DRPC – HTTP / WebSocket
- Access: `eth_getLogs`, `eth_call`, `eth_subscribe` or libraries `web3py`, `go-ethereum`
- **Why raw alone is not enough:** query "all swaps of user X for a year" = scanning millions of blocks. You need **indexes**.

The Graph – Indexing via Subgraphs

Idea: developers define how to transform on-chain events into a structured schema queryable via GraphQL.



The Graph – Query Example

```
# Top 5 Uniswap V3 pools by TVL
{
  pools(first: 5, orderBy: totalValueLockedUSD, orderDirection: desc) {
    id
    token0 { symbol }
    token1 { symbol }
    feeTier
    totalValueLockedUSD
    volumeUSD
  }
}
```

Pros: fast iteration, community subgraphs for most major protocols, open-source.
Cons: GraphQL limits on complex joins / aggregations; sync lag; no cross-subgraph joins.

More Indexers

- **Goldsky** – *streaming / mirroring*. Instead of GraphQL, pipe on-chain data directly into your Postgres / Kafka / ClickHouse. Low latency, good for custom backends.
- **Bitquery** – *multi-chain GraphQL + REST*. Strong cross-chain coverage (40+ chains), DEX trades, token transfers out of the box. Good for research and bots.
- **Space and Time** – *SQL + verifiability*. Decentralized warehouse with zero-knowledge proofs of query correctness. Target: smart contracts that need off-chain compute.

How to choose:

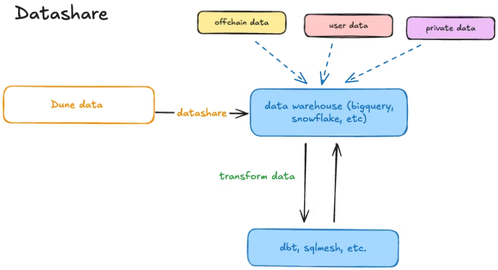
- Need a public API for your dApp? → **The Graph**
- Need low-latency streams into your own DB? → **Goldsky**
- Need multi-chain research queries? → **Bitquery**
- Need on-chain verifiable analytics? → **Space and Time**

Scanners

- User-friendly transactions explore: Etherscan, Arbiscan, Bscscan, Solscan
- State explore: human-readable view of transactions, contracts, tokens
- API: contract ABIs, verified source, token balances, decoded events and indexed on-chain data with fast access
- Not an analytics platform: lacks joins, dashboards, complex aggregation

Dune Analytics

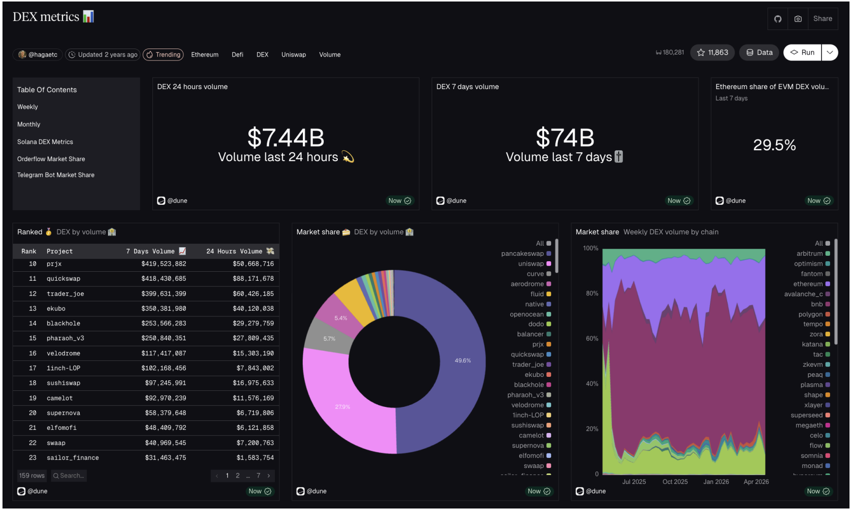
- Decoded tables on top of raw blockchain data (e.g. `uniswap_v3.swaps`)
- SQL interface (DuneSQL, Trino-based), community dashboards, forks
- Multichain: Ethereum, Arbitrum, Optimism, Base, Solana, etc.
- Model: "analyst as author" – open dashboards, reusable queries, access to API, custom indexing



Dune SQL Example

```sql
-- Daily volume of a specific Uniswap V3 pool
SELECT
  date_trunc('day', block_time) AS day,
  SUM(amount_usd) AS volume_usd
FROM uniswap_v3_ethereum.trades
WHERE project_contract_address = 0x88e6a0c2ddd26feeb64f039a2c41296fcb3f5640 -- USDC/ETH 0.05%
  AND block_time > NOW() - INTERVAL '30' DAY
GROUP BY 1
ORDER BY 1 DESC;
```

Dune Dashboard Example



Analytics

Defillama:

- Specializes in TVL across 3000+ protocols and 300+ chains
- Open methodology, open-source adapters on GitHub for fast protocols
- Also covers stablecoins, yields, fees, DEX volumes
- Open-accessed API with a history

Dexscreener / GeckoTerminal:

- Real-time DEX pair charts, trending, new launches
- Used heavily by memecoin / sniper communities
- Reports: volume, liquidity, buys vs sells, holder count

Market Data

Centralized exchanges (CEXs):

- Binance API – REST + WebSocket; order book, trades, OHLCV (klines)
- Other CEX with public APIs: OKX, Bybit, Coinbase, Kraken
- Latency: WebSocket ~tens of ms; REST ~hundreds of ms

Crypto aggregators:

- **CoinGecko, CoinMarketCap** – cross-exchange price aggregation, market caps
- Available markets and meta data
- The unified crypto data
- Important: their prices are often averaged with volume weighting

Oracles – the Off-chain to On-chain Bridge

Problem: smart contracts are deterministic and cannot call external APIs. But DeFi needs prices, randomness, real-world events.

- Push-based (Chainlink feeds): off-chain aggregator pushes updates on-chain on a heartbeat or deviation threshold. Gas is paid by the oracle network. Simple for consumers: just read a contract.
- Pull-based (Pyth or Data Streams): data lives off-chain in Pythnet. Consumer pulls a signed update and submits it on-chain when needed. User pays gas. Lower latency, better for high-frequency protocols.

AI $\times$ Blockchain Data – Use Cases

- Anomaly and fraud detection – suspicious transactions, wash trading, sybil clusters, rug-pull patterns
- On-chain intelligence – wallet clustering, identity resolution, smart money tracking (core to Nansen, Arkham, ChainAlysis)
- Market forecasting, algorithmic trading and liquidity Management – autonomously read market data, events, and execute on-chain transactions
- Decentralized infrastructure for AI models: decentralized training and compute (federated learning), verifiable AI with decentralized inference, inference scoring and aggregation

Summary

- Blockchain data is huge and heterogeneous – hundreds of TB and growing
- No single tool solves every problem – different levels of aggregation exist for different use cases
- Choice of level = trade-off between *control, cost, latency, trust*

References I

- Etherscan — Ethereum Block Explorer
- The Graph — Documentation and Subgraph Studio
- Goldsky — Real-time blockchain data pipelines
- Bitquery — Multi-chain GraphQL API
- Space and Time — Verifiable data warehouse
- Dune Analytics — SQL-based blockchain analytics
- DefiLlama — TVL and DeFi metrics
- Dexscreener — DEX pair charts and trending

References II

- GeckoTerminal — DEX analytics by CoinGecko
- Chainlink — Price Feeds documentation
- Pyth Network — Pull-based oracle documentation
- Nansen — On-chain intelligence platform